THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 1	Digit sum must be a multiple of 3.
2	**(c)** 4	Count the terms.
3	**(d)** 6174	Largest arrangement minus smallest arrangement.
4	**(d)** 9	The first half decides the number.
5	**(d)** 60	Use the middle number times the count when the count is odd.
6	**(a)** even	Test with small values of n.
7	**(a)** 3 but not 9	Use the digit-sum tests.
8	**(b)** 99	Reverse the digits, then add.
9	**(a)** 2	Digit sum must be a multiple of 3.
10	**(b)** 15	Count the terms.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Digit sum $= 10 + *$; Must be a multiple of 9, so $* = 8$ (1); Number $= 5814$ (1).	1+1
12	Middle odd number $= 51 \div 3 = 17$; Others differ by 2: 15 and 19 (1); Numbers: $15, 17, 19$; Check sum $= 51$ (1).	1+1
13	$39 + 93 = 132$; $132 + 231 = 363$ (1); Steps $= 2$; palindrome 363 (1).	1+1
14	Digit sum $= 15 + *$; Must be a multiple of 9, so $* = 3$ (1); Number $= 6237$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$7621 - 1267 = 6354$; $6543 - 3456 = 3087$; $8730 - 0378 = 8352$; $8532 - 2358 = 6174$ (1); Steps needed $= 4$ (2).	1+2
16	$11(a + b) = 99$, so $a + b = 9$; $a - b = 3$ (1); $a = 6$, $b = 3$ (1); Number $= 63$ (1).	1+1+1
17	Middle number $= 75 \div 5 = 15$; First number $= 15 - 2 = 13$ (1); Numbers: $13, 14, 15, 16, 17$ (1); Check sum $= 75$ (1).	1+1+1
	OR: $9810 - 0189 = 9621$; $9621 - 1269 = 8352$; $8532 - 2358 = 6174$ (1); Steps needed $= 3$ (2).	1+2
18	$11(a + b) = 121$, so $a + b = 11$; $a - b = 1$ (1); $a = 6$, $b = 5$ (1); Number $= 65$ (1).	1+1+1
	OR: Middle number $= 63 \div 7 = 9$; First number $= 9 - 3 = 6$ (1); Numbers: $6, 7, 8, 9, 10, 11, 12$ (1); Check sum $= 63$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sequence: $7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Largest $= 52$ (1); Even terms $= 11$ (1); Odd terms $= 6$ (1); Total terms $= 17$ (1).	1+1+1+1+1
	OR: Palindrome form $1b1$ with digit sum $= 2 \times 1 + b$ (1); Need $2 + b$ to be a multiple of 3 (1); Possible b: $1, 4, 7$ (1); Palindromes: $111, 141, 171$ (2).	1+1+1+2
20	Sequence: $20, 10, 5, 16, 8, 4, 2, 1$ (1); Largest $= 20$ (1); Even terms $= 6$ (1); Odd terms $= 2$ (1); Total terms $= 8$ (1).	1+1+1+1+1
	OR: Palindrome form $2b2$ with digit sum $= 2 \times 2 + b$ (1); Need $4 + b$ to be a multiple of 3 (1); Possible b: $2, 5, 8$ (1); Palindromes: $222, 252, 282$ (2).	1+1+1+2

Mistakes to avoid:

- Saying the sum of two odd numbers is odd; it is even.
- Treating the product of two odd numbers as even; it is odd.
- Writing the reverse of $10a + b$ as $10a + b$ again; the reverse is $10b + a$.
- Using the divisibility test for 9 as if it were for 3 (digit sum 12 is divisible by 3 but not 9).
- In Kaprekar's routine, forgetting to keep leading zeros when the smaller arrangement starts with 0.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 9261	Largest arrangement minus smallest arrangement.
2	**(a)** 90	The first half decides the number.
3	**(d)** 66	Use the middle number times the count when the count is odd.
4	**(b)** odd	Test with small values of n.
5	**(a)** both 3 and 9	Use the digit-sum tests.
6	**(b)** 968	Reverse the digits, then add.
7	**(a)** 2	Digit sum must be a multiple of 3.
8	**(c)** 18	Count the terms.
9	**(a)** 8352	Largest arrangement minus smallest arrangement.
10	**(d)** 90	The first half decides the number.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Middle odd number $= 87 \div 3 = 29$; Others differ by 2: 27 and 31 (1); Numbers: $27, 29, 31$; Check sum $= 87$ (1).	1+1
12	$58 + 85 = 143$; $143 + 341 = 484$ (1); Steps $= 2$; palindrome 484 (1).	1+1
13	Digit sum $= 17 + *$; Must be a multiple of 9, so $* = 1$ (1); Number $= 3186$ (1).	1+1
14	Middle odd number $= 117 \div 3 = 39$; Others differ by 2: 37 and 41 (1); Numbers: $37, 39, 41$; Check sum $= 117$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$5432 - 2345 = 3087$; $8730 - 0378 = 8352$; $8532 - 2358 = 6174$ (1); Steps needed $= 3$ (2).	1+2
16	$11(a + b) = 110$, so $a + b = 10$; $a - b = 4$ (1); $a = 7$, $b = 3$ (1); Number $= 73$ (1).	1+1+1
17	Middle number $= 99 \div 9 = 11$; First number $= 11 - 4 = 7$ (1); Numbers: $7, 8, 9, 10, 11, 12, 13, 14, 15$ (1); Check sum $= 99$ (1).	1+1+1
	OR: $8721 - 1278 = 7443$; $7443 - 3447 = 3996$; $9963 - 3699 = 6264$; $6642 - 2466 = 4176$; $7641 - 1467 = 6174$ (1); Steps needed $= 5$ (2).	1+2
18	$11(a + b) = 143$, so $a + b = 13$; $a - b = 5$ (1); $a = 9$, $b = 4$ (1); Number $= 94$ (1).	1+1+1
	OR: Middle number $= 84 \div 3 = 28$; First number $= 28 - 1 = 27$ (1); Numbers: $27, 28, 29$ (1); Check sum $= 84$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sequence: $21, 64, 32, 16, 8, 4, 2, 1$ (1); Largest $= 64$ (1); Even terms $= 6$ (1); Odd terms $= 2$ (1); Total terms $= 8$ (1).	1+1+1+1+1
	OR: Palindrome form $4b4$ with digit sum $= 2 \times 4 + b$ (1); Need $8 + b$ to be a multiple of 3 (1); Possible b: $1, 4, 7$ (1); Palindromes: $414, 444, 474$ (2).	1+1+1+2
20	Sequence: $22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Largest $= 52$ (1); Even terms $= 11$ (1); Odd terms $= 5$ (1); Total terms $= 16$ (1).	1+1+1+1+1
	OR: Palindrome form $5b5$ with digit sum $= 2 \times 5 + b$ (1); Need $10 + b$ to be a multiple of 3 (1); Possible b: $2, 5, 8$ (1); Palindromes: $525, 555, 585$ (2).	1+1+1+2

Mistakes to avoid:

- Saying the sum of two odd numbers is odd; it is even.
- Treating the product of two odd numbers as even; it is odd.
- Writing the reverse of $10a + b$ as $10a + b$ again; the reverse is $10b + a$.
- Using the divisibility test for 9 as if it were for 3 (digit sum 12 is divisible by 3 but not 9).
- In Kaprekar's routine, forgetting to keep leading zeros when the smaller arrangement starts with 0.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 133	Use the middle number times the count when the count is odd.
2	**(b)** odd	Test with small values of n.
3	**(a)** neither 3 nor 9	Use the digit-sum tests.
4	**(d)** 10230	Reverse the digits, then add.
5	**(d)** 1	Digit sum must be a multiple of 3.
6	**(a)** 5	Count the terms.
7	**(a)** 7083	Largest arrangement minus smallest arrangement.
8	**(d)** 900	The first half decides the number.
9	**(b)** 99	Use the middle number times the count when the count is odd.
10	**(b)** even but not a multiple of 4	Test with small values of n.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$153 + 351 = 504$; $504 + 405 = 909$ (1); Steps $= 2$; palindrome 909 (1).	1+1
12	Digit sum $= 15 + *$; Must be a multiple of 9, so $* = 3$ (1); Number $= 8136$ (1).	1+1
13	Middle odd number $= 159 \div 3 = 53$; Others differ by 2: 51 and 55 (1); Numbers: $51, 53, 55$; Check sum $= 159$ (1).	1+1
14	$162 + 261 = 423$; $423 + 324 = 747$ (1); Steps $= 2$; palindrome 747 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$8531 - 1358 = 7173$; $7731 - 1377 = 6354$; $6543 - 3456 = 3087$; $8730 - 0378 = 8352$; $8532 - 2358 = 6174$ (1); Steps needed $= 5$ (2).	1+2
16	$11(a + b) = 88$, so $a + b = 8$; $a - b = 6$ (1); $a = 7$, $b = 1$ (1); Number $= 71$ (1).	1+1+1
17	Middle number $= 130 \div 5 = 26$; First number $= 26 - 2 = 24$ (1); Numbers: $24, 25, 26, 27, 28$ (1); Check sum $= 130$ (1).	1+1+1
	OR: $8542 - 2458 = 6084$; $8640 - 0468 = 8172$; $8721 - 1278 = 7443$; $7443 - 3447 = 3996$; $9963 - 3699 = 6264$; $6642 - 2466 = 4176$; $7641 - 1467 = 6174$ (1); Steps needed $= 7$ (2).	1+2
18	$11(a + b) = 77$, so $a + b = 7$; $a - b = 5$ (1); $a = 6$, $b = 1$ (1); Number $= 61$ (1).	1+1+1
	OR: Middle number $= 105 \div 7 = 15$; First number $= 15 - 3 = 12$ (1); Numbers: $12, 13, 14, 15, 16, 17, 18$ (1); Check sum $= 105$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sequence: $24, 12, 6, 3, 10, 5, 16, 8, 4, 2, 1$ (1); Largest $= 24$ (1); Even terms $= 8$ (1); Odd terms $= 3$ (1); Total terms $= 11$ (1).	1+1+1+1+1
	OR: Palindrome form $7b7$ with digit sum $= 2 \times 7 + b$ (1); Need $14 + b$ to be a multiple of 3 (1); Possible b: $1, 4, 7$ (1); Palindromes: $717, 747, 777$ (2).	1+1+1+2
20	Sequence: $18, 9, 28, 14, 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Largest $= 52$ (1); Even terms $= 14$ (1); Odd terms $= 7$ (1); Total terms $= 21$ (1).	1+1+1+1+1
	OR: Palindrome form $8b8$ with digit sum $= 2 \times 8 + b$ (1); Need $16 + b$ to be a multiple of 3 (1); Possible b: $2, 5, 8$ (1); Palindromes: $828, 858, 888$ (2).	1+1+1+2

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